What QTLs are affecting Kernel Discoloration Resistance and Grain Protein Content in Spring Malting Barley (Hordeum vulgare L.)?

Cameron Matthews | North Dakota State University

Factors impacting kernel discoloration

- Biotic Factors:
  - Bacteria/fungi presence
  - Genetics
- Abiotic Factors:
  - High humidity
  - Heavy rain fall
  - Extreme temperatures

Genetics Controlling Low-Protein Characteristics

- The “low-protein gene” used in many programs is from the cultivar ‘Karl.’
- Cultivar was developed by USDA-ARS in Aberdeen, ID
- This low-protein gene is located in chromosome 6H
- This low-protein gene is responsible for decreased total protein content and hordein accumulation in the endosperm.
- The gene reduces grain protein by 1.0 to 1.5 percentage points.
- Lines with the low-protein gene have also been reported to have poor kernel color and low diastatic power (DP).

Climate Comparison of Growing Regions

Climate Data from U.S. Climate Data

Fargo, North Dakota During Grain Fill:

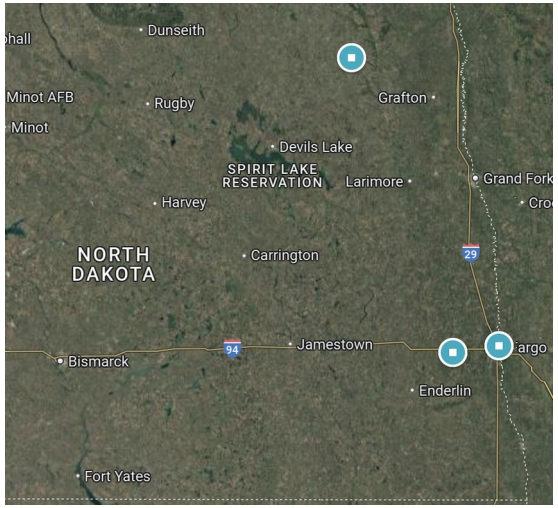
<u>Average</u>

Low Temperature: 15.6 ˚C

High Temperature: 28.3 ˚C

Precipitation: 70.9 mm

Altitude: 273 m



Aberdeen, Idaho During Grain Fill:

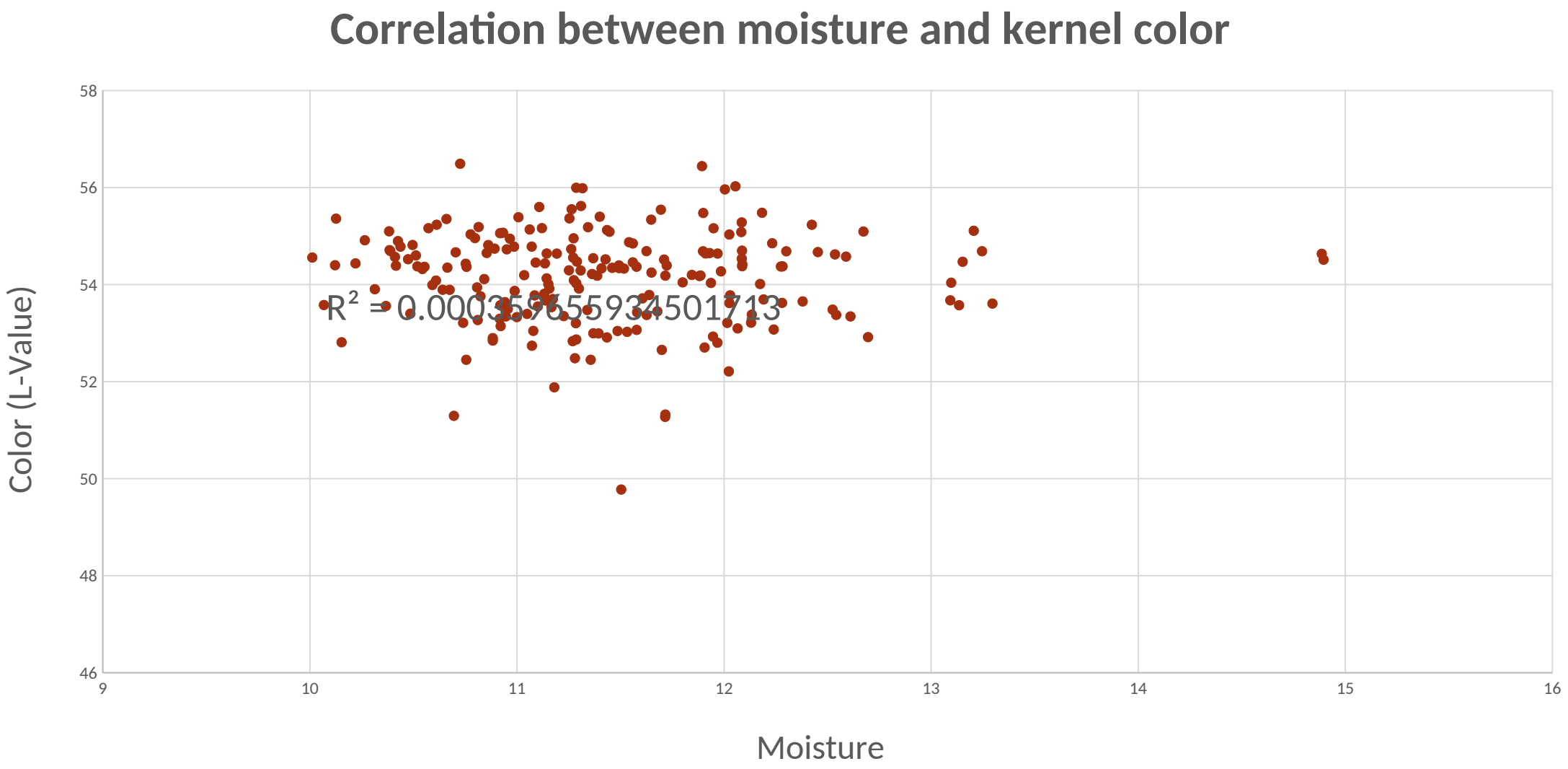
<u>Average</u>

Low Temperature: 10.0 ˚C

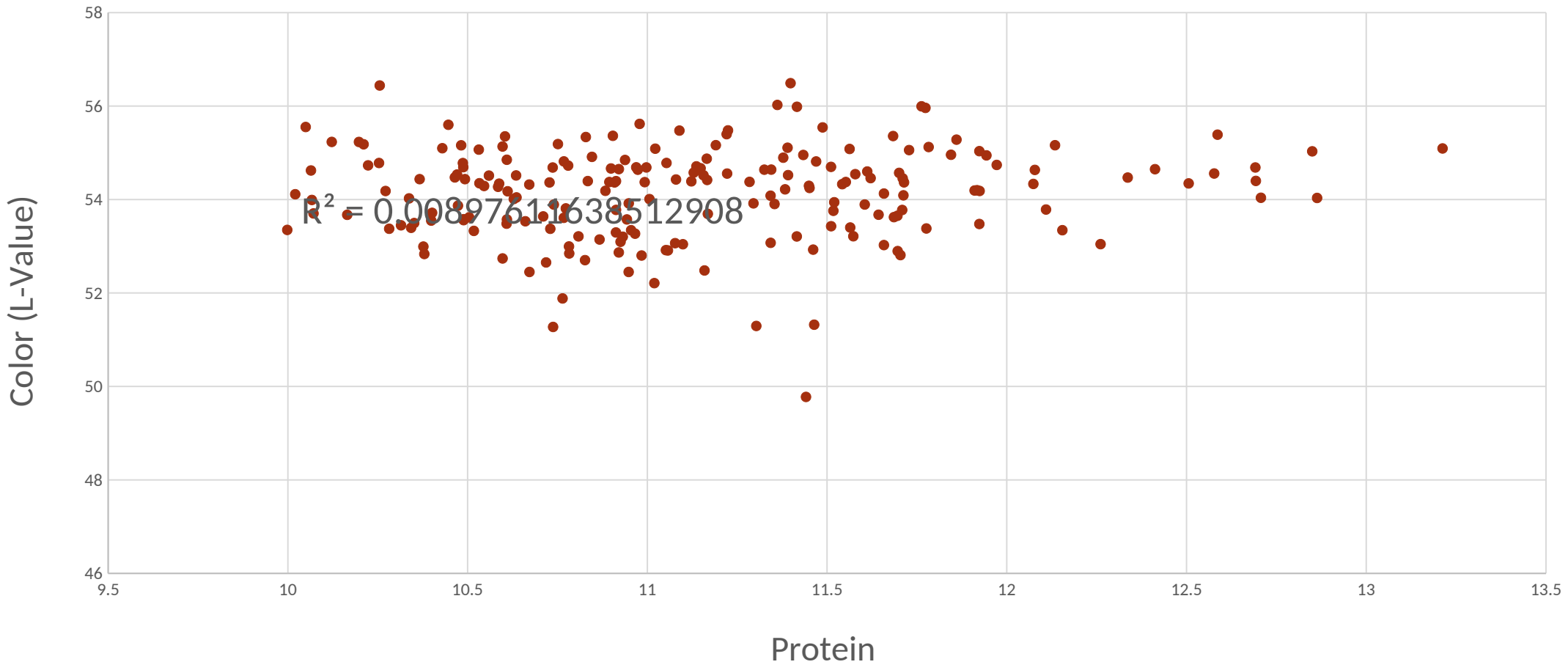
High Temperature: 30.4 ˚C

Precipitation: 13 mm

Altitude: 1358 m


Correlation between moisture and kernel color
Color (L-Value)
Moisture
R² = 0.00035965593450171 3
58
56
54
52
50
48
46
9
10
11
12
13
14
15
16

Correlation between protein and kernel color



Is Kernel Discoloration Resistance Correlated With Other Malt Quality Traits?

Correlations

	PLUMP	PROTEIN	COLOR	KWT	M_EXTRACT	SOVERT	WRTPRT	W_CLARITY	W_COLOR	A_AMYLASE	DP	B_GLUCAN	FAN
PLUMP	1.0000	-0.0328	-0.1449	0.5711	0.1977	0.1202	0.0689	-0.0277	0.0310	0.0170	-0.0846	0.0994	0.0557
PROTEIN	-0.0328	1.0000	-0.0343	-0.0250	-0.2571	-0.2650	0.1342	-0.0142	-0.0151	-0.0812	0.2253	0.0372	0.0937
COLOR	-0.1449	-0.0343	1.0000	-0.1451	0.0429	0.0642	0.0221	0.0660	-0.0240	-0.0279	0.0152	-0.0531	0.0291
KWT	0.5711	-0.0250	-0.1451	1.0000	0.1701	0.1258	0.1045	-0.0382	0.0580	-0.0393	-0.1085	0.0230	0.1090
M_EXTRACT	0.1977	-0.2571	0.0429	0.1701	1.0000	0.5480	0.2581	0.0122	0.1088	0.3069	-0.2131	-0.1173	0.2779
SOVERT	0.1202	-0.2650	0.0642	0.1258	0.5480	1.0000	0.7061	0.1002	0.2937	0.4295	-0.1258	-0.3514	0.6024
WRTPRT	0.0689	0.1342	0.0221	0.1045	0.2581	0.7061	1.0000	0.0811	0.3214	0.3647	0.1493	-0.3733	0.7765
W_CLARITY	-0.0277	-0.0142	0.0660	-0.0382	0.0122	0.1002	0.0811	1.0000	0.2369	0.0498	0.0194	-0.1462	0.1079
W_COLOR	0.0310	-0.0151	-0.0240	0.0580	0.1088	0.2937	0.3214	0.2369	1.0000	0.1752	-0.0681	-0.1868	0.3186
A_AMYLASE	0.0170	-0.0812	-0.0279	-0.0393	0.3069	0.4295	0.3647	0.0498	0.1752	1.0000	0.1329	-0.2407	0.3918
DP	-0.0846	0.2253	0.0152	-0.1085	-0.2131	-0.1258	0.1493	0.0194	-0.0681	0.1329	1.0000	-0.2055	0.0979
B_GLUCAN	0.0994	0.0372	-0.0531	0.0230	-0.1173	-0.3514	-0.3733	-0.1462	-0.1868	-0.2407	-0.2055	1.0000	-0.4089
FAN	0.0557	0.0937	0.0291	0.1090	0.2779	0.6024	0.7765	0.1079	0.3186	0.3918	0.0979	-0.4089	1.0000

There are 112 missing values. The correlations are estimated by Pairwise method.

Characterizing Kernel Discoloration

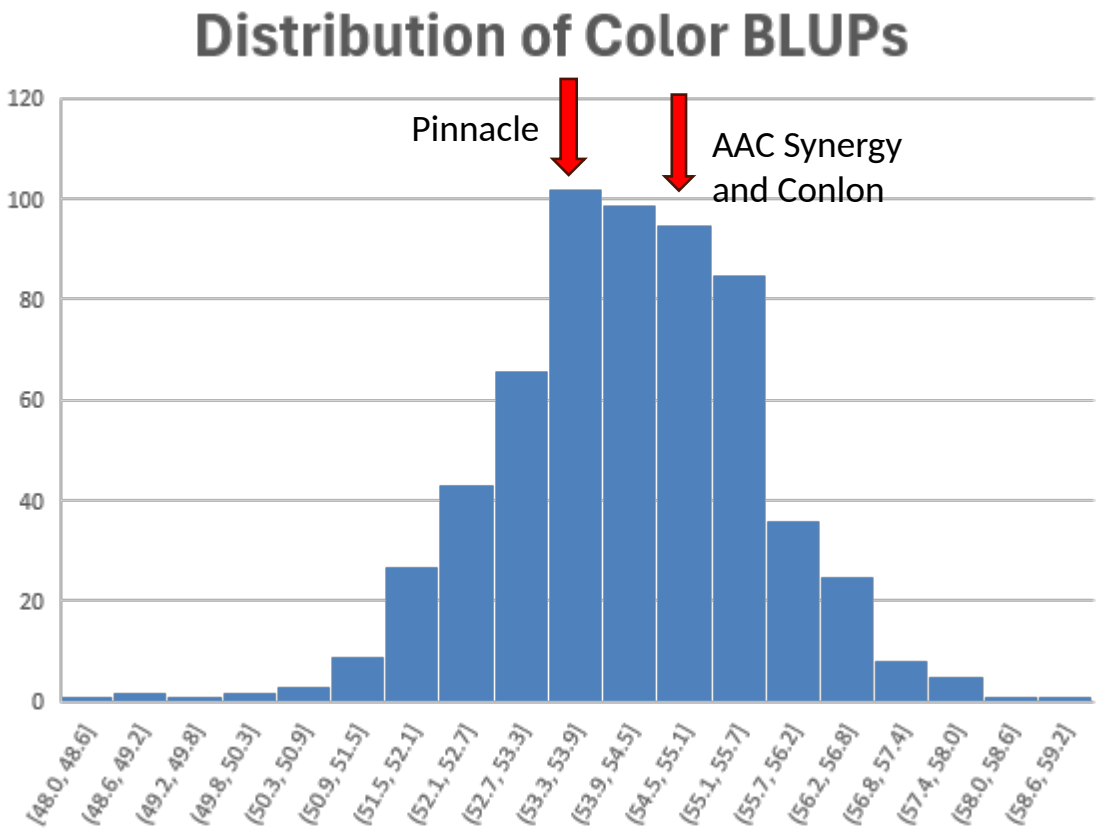
- Genotypes were grown in an augmented block design with checks arranged on the diagonal at multiple locations in 2023 and 2024
- Each entry was genotyped using the 50K Illumina Infinium iSelect array for barley

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13
13	100	1	154	19	97	11	102	46	52	40	2	138	55
12	141	38	94	84	42	4	68	127	77	63	53	153	129
11	2	108	113	104	74	27	144	79	76	3	35	142	43
10	6	148	50	28	2	26	37	85	12	87	47	20	65
9	75	143	33	44	93	64	132	17	2	56	95	88	41
8	48	139	107	4	31	131	126	70	62	13	96	81	1
7	10	122	117	99	25	118	30	1	101	136	133	115	106
6	120	15	3	5	135	8	73	36	59	51	121	1	112
5	60	110	145	14	123	119	4	18	71	66	151	80	58
4	91	3	67	78	22	57	83	140	34	89	1	116	69
3	103	54	137	82	9	4	149	134	72	23	130	105	32
2	3	29	98	86	146	124	147	90	16	2	109	92	61
1	114	49	7	45	4	128	111	24	152	39	125	21	150

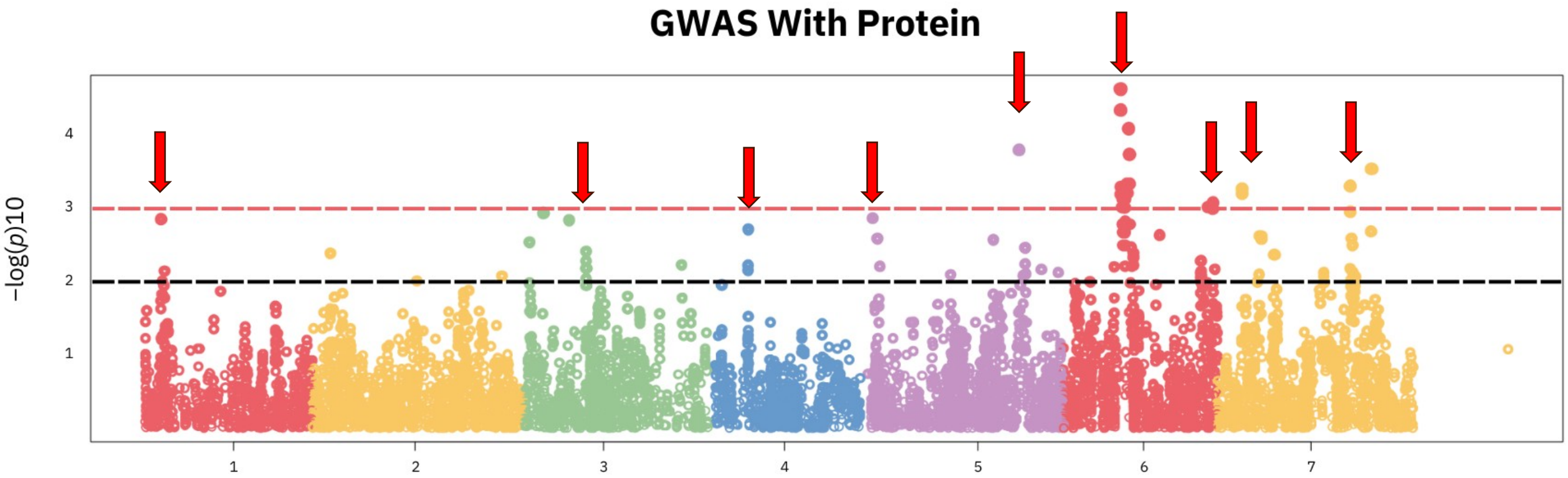
(Murillo Florez et. al, 2021)

Characterizing Kernel Discoloration

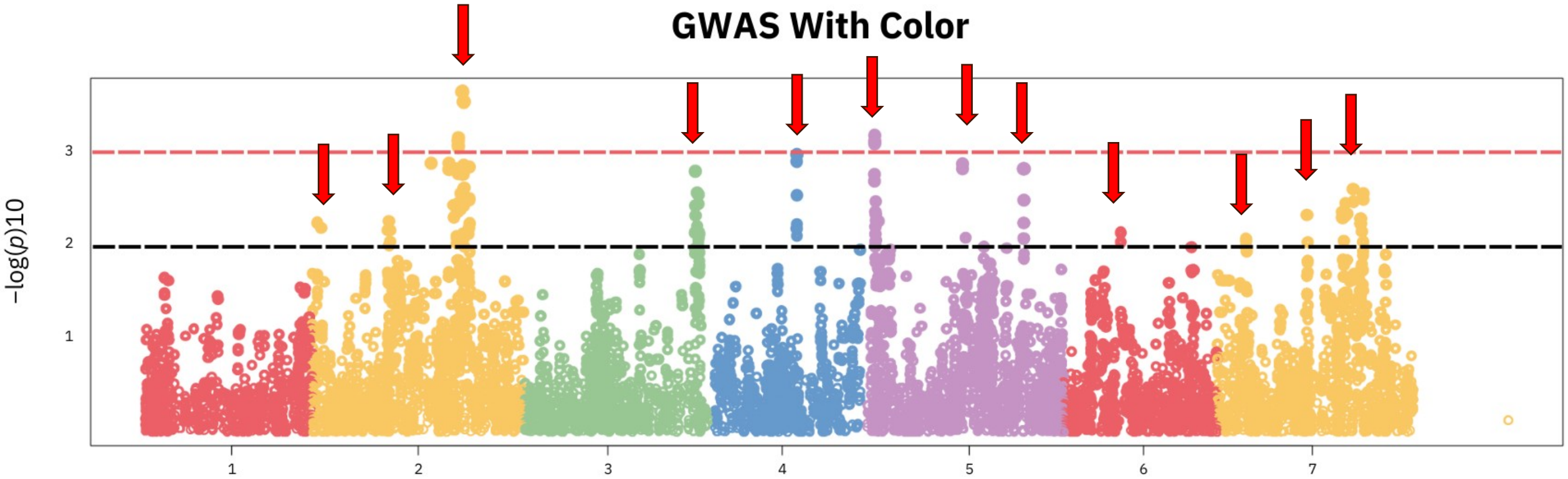
- Each genotype was tested for protein content using an NIR and for brightness using a color spectrometer.
- Best Linear Unbiased Estimators (BLUEs) for single locations and Best Linear Unbiased Predictors (BLUPs) across locations were calculated and used for GWAS analysis to identify significant QTLs



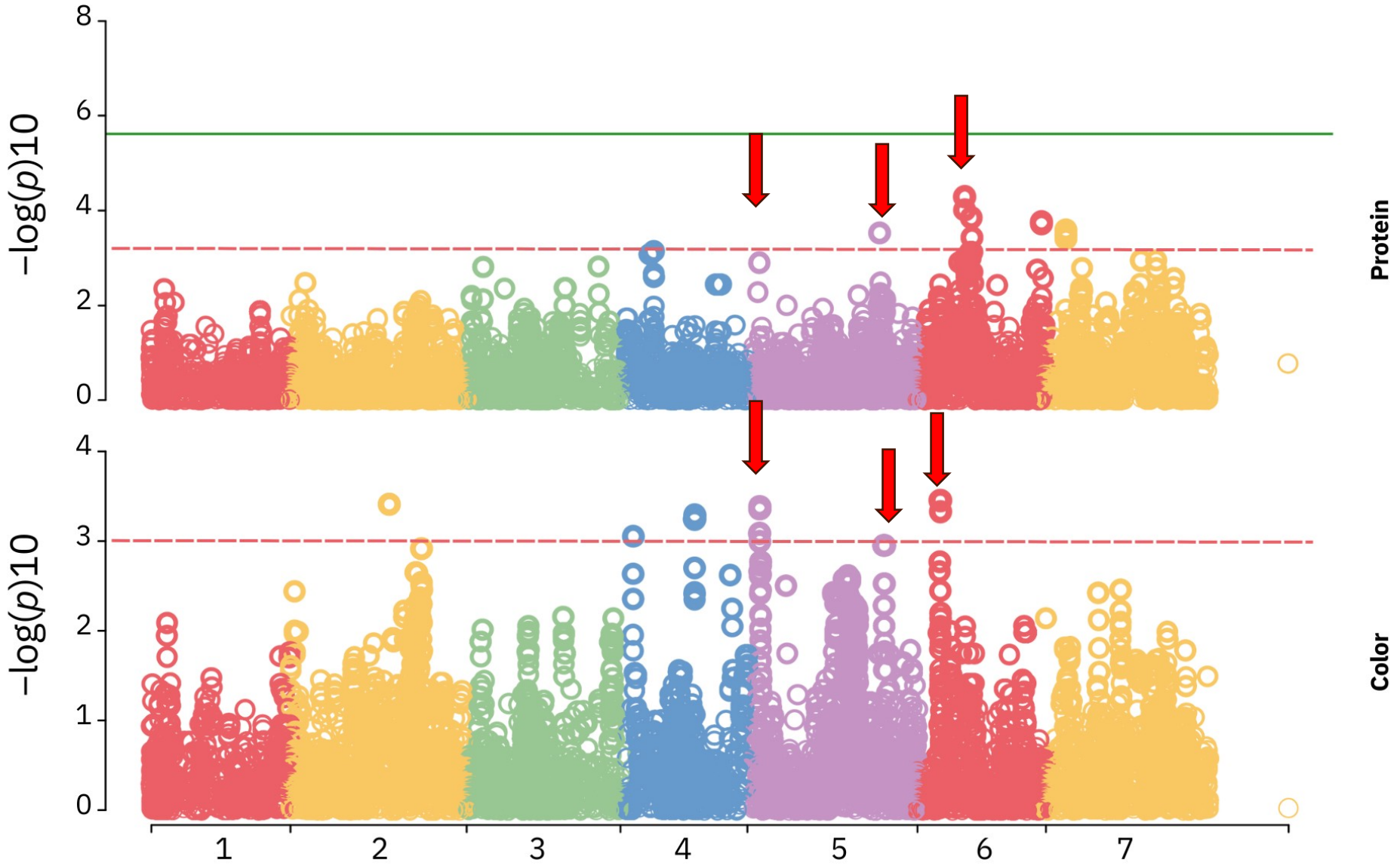
Protein GWAS



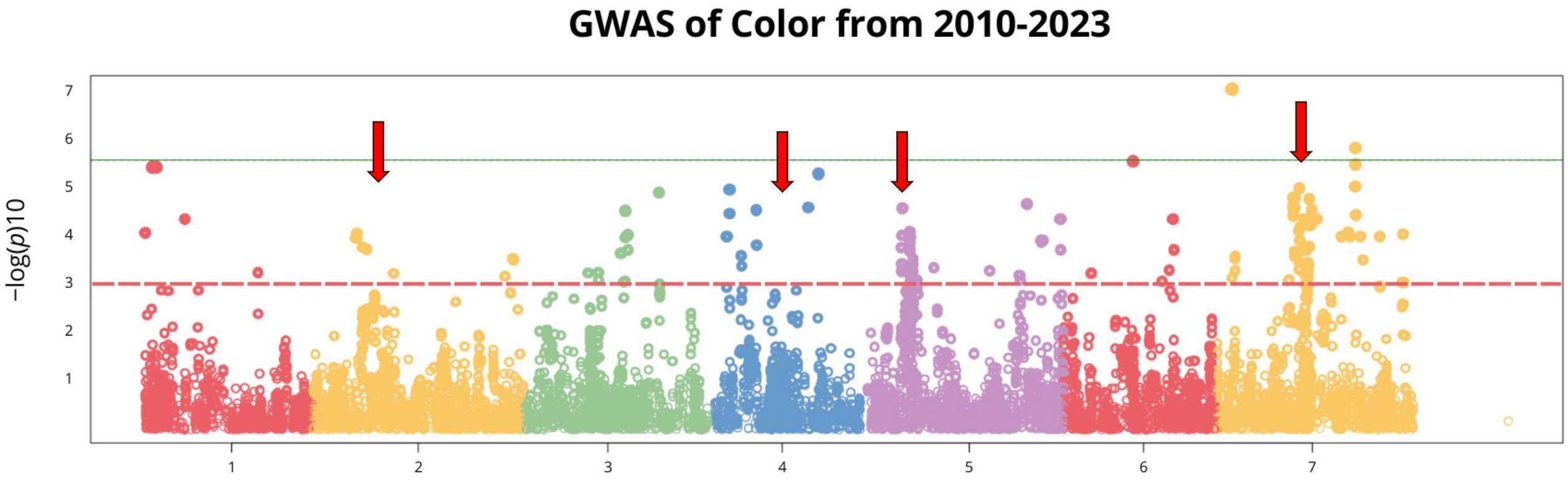
Color GWAS



Comparison of Protein and Color GWAS



GWAS of Color Across All Available Years



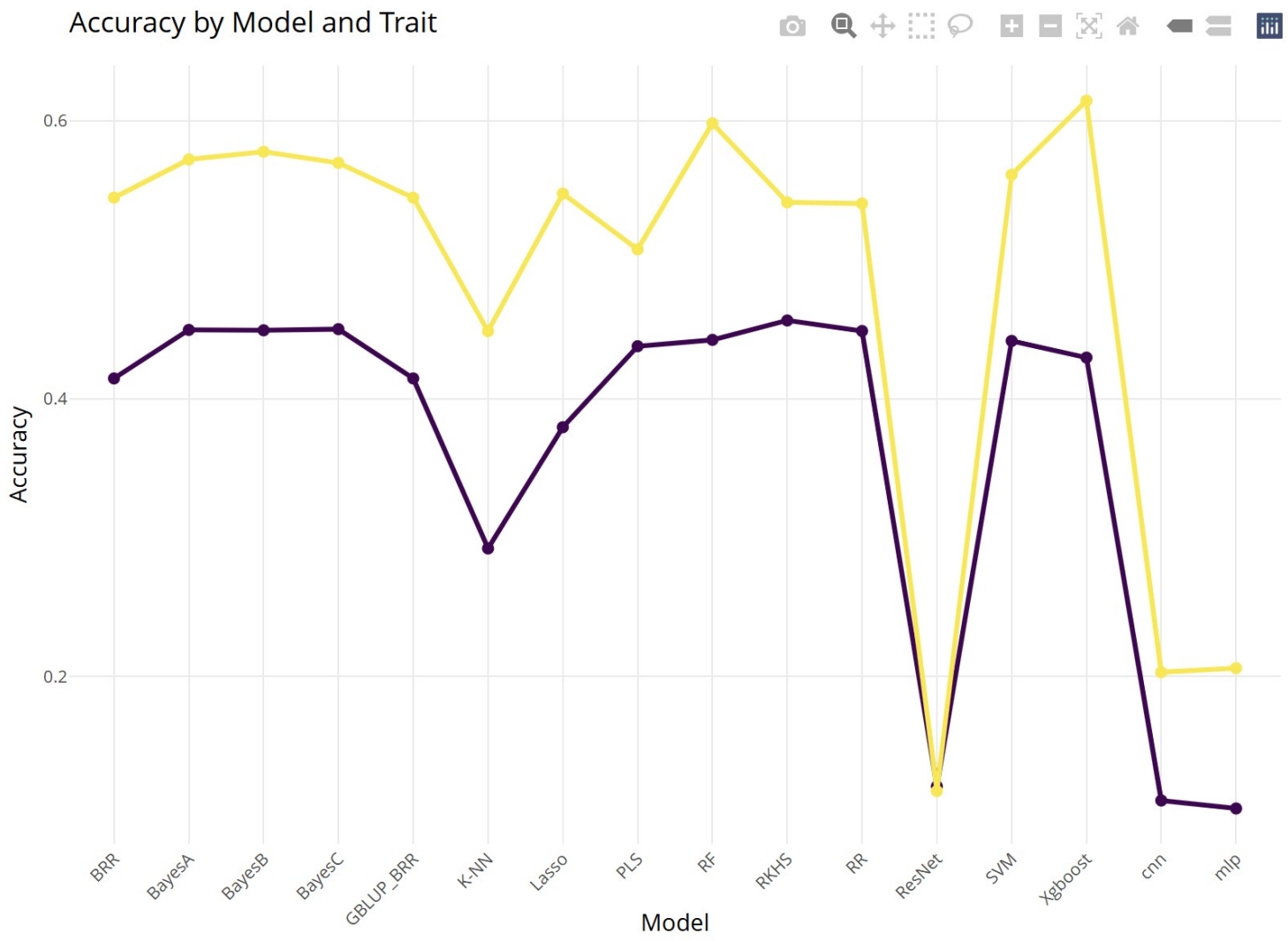
Filtering for Lines With Acceptable Performance on All Traits

Filters	Total Genotype Count
Starting	2708
After Malt Extract	1376
After Beta Glucan	842
After Diastatic Power	395
After Alpha Amylase	270
After Plump	170
After Protein	103
After Free Amino Nitrogen	67
After Wort Protein	52
After SOVERT	50
After Wort Color	7
After Kernel Weight	5
After Kernel Color	2

Current Genomic Selection Models Based On The NDSU Designed 3.6K SNP ChIP

Trait	CVMC correlation	Reliability diagram comments	Method
Alpha-amylase	0.632	Good ranking of lines throughout	RR
Beta-glucan	0.753	Good ranking of lines throughout	RR
Carbohydrate extract	0.446	Unreliable for low and high values	RR
DP	0.596	Good ranking of lines throughout	RR
DP/Protein	0.620	Good ranking of lines for identifying low values	RR
Extract	0.451	OK ranking of lines throughout	RR
FAN	0.589	Good ranking of lines throughout	RR
Heading date	0.717	Good ranking except at the very top two points	RR
Height	0.489	Good ranking of lines throughout	RR
Protein	0.590	Ok ranking of lines except at top and bottom	RR
S/T	0.560	Good ranking of lines throughout	RR
Turbidity	0.756	Good ranking except at the very top two points	RR
Viscosity	0.694	Good ranking except for the very last bottom points	RR
Wort color opt	0.328	Good ranking throughout, but the top two points are off the line	RR
Wort protein	0.614	Good ranking throughout, but bottom point is off the line	RR
Yield	0.326	Good ranking at the top but problematic at the last four points	RR

Development of a Genomic Selection Model for Kernel Discoloration Resistance and Grain Protein Content Using The 50K Illumina Infinium iSelect ChIP



Show 12 entries

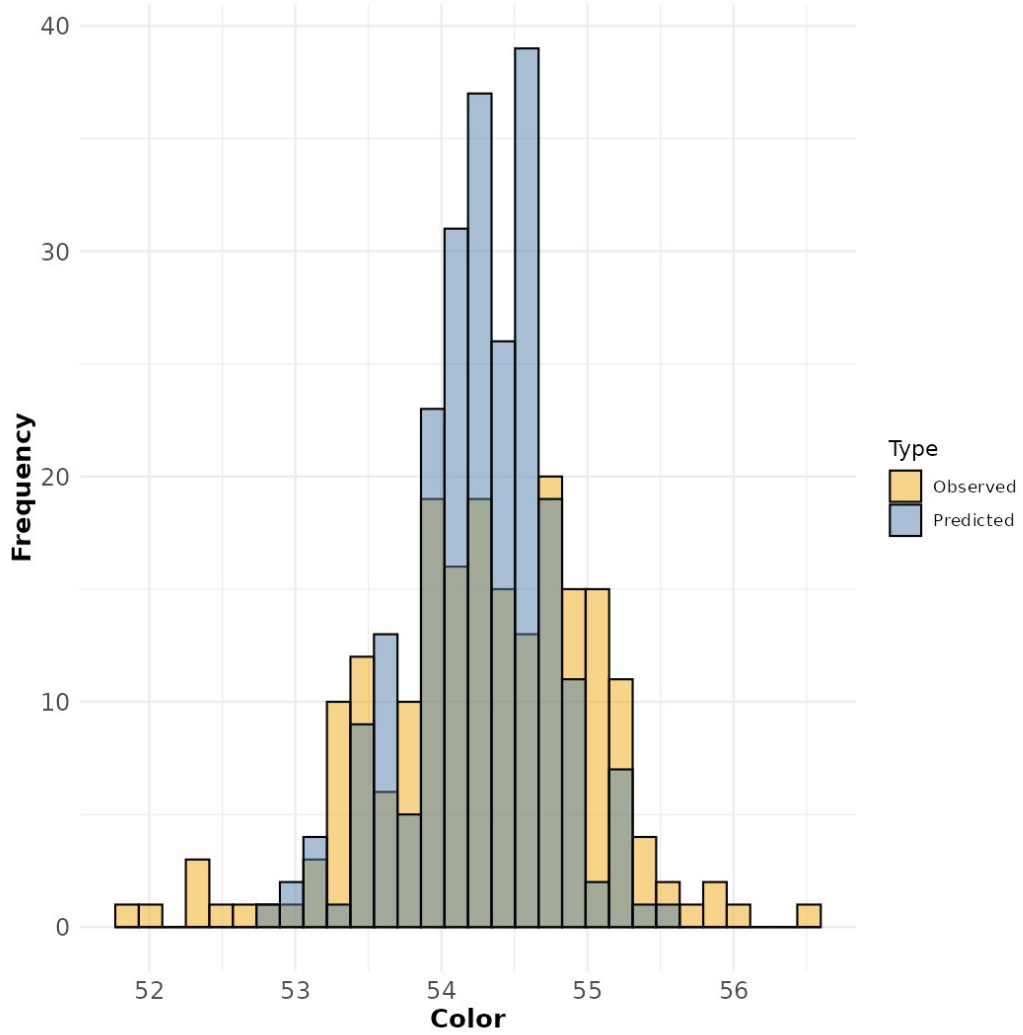
Best models for each trait:

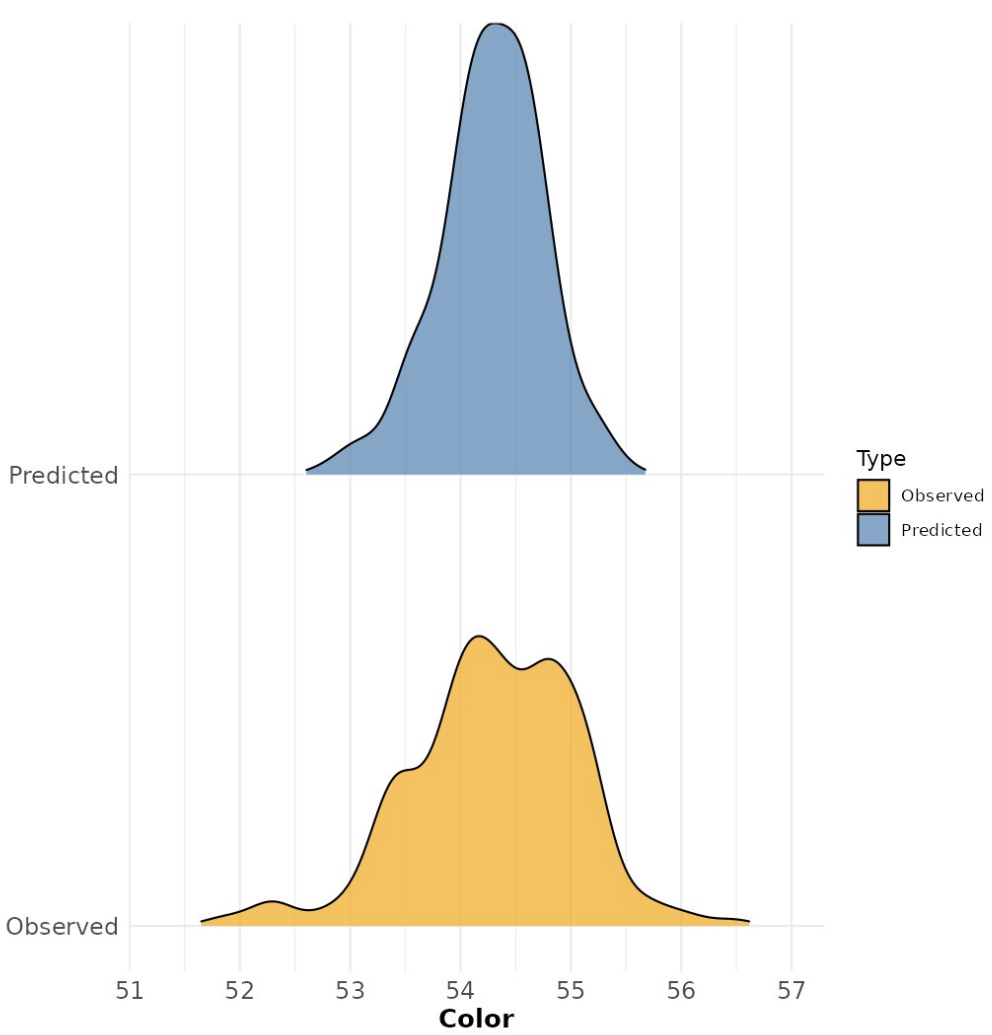
trait	model	accuracy
Protein	Xgboost	0.6145
Color	RKHS	0.4563

Showing 1 to 2 of 2 entries

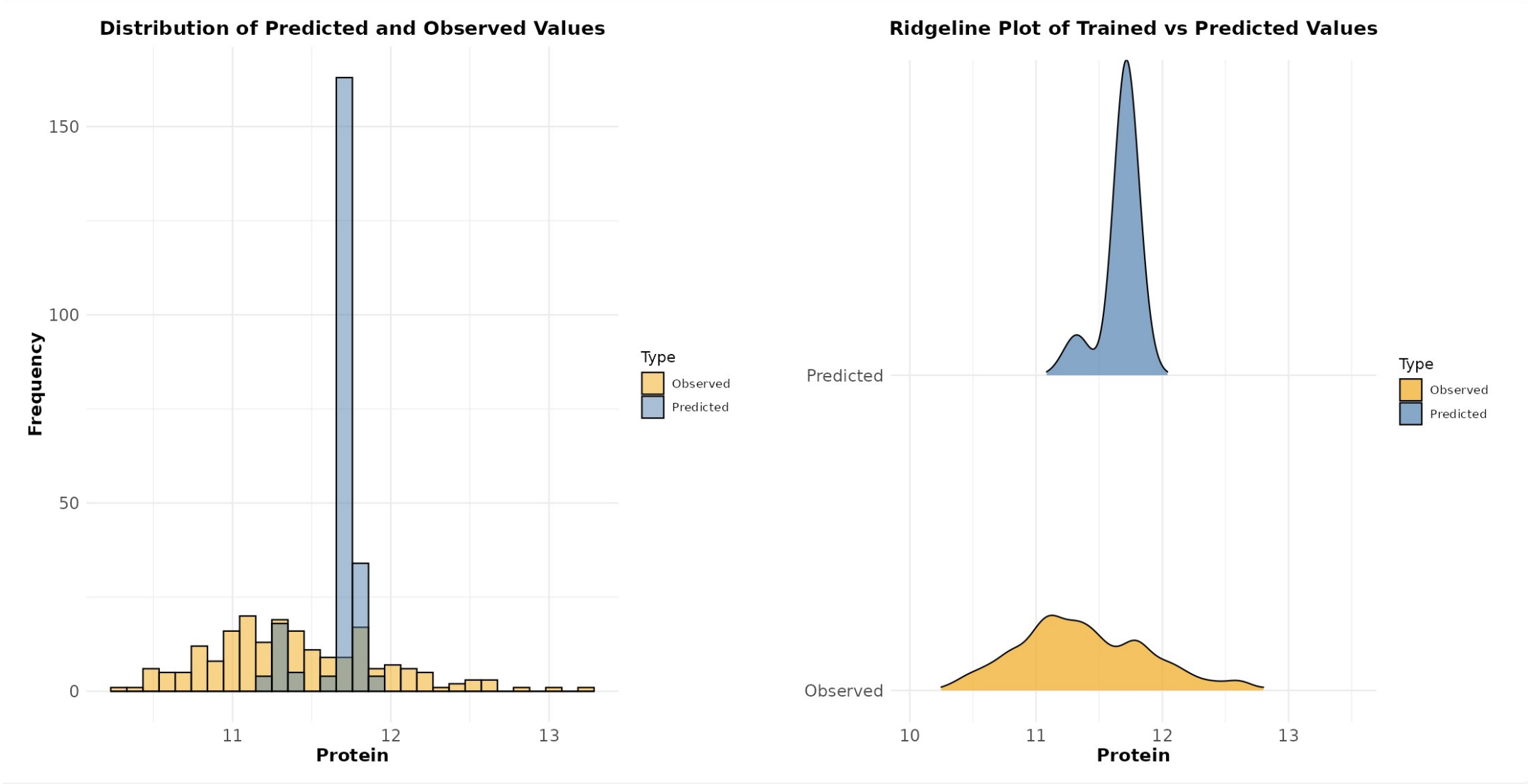
Previous 1 Next

Distribution of Observed vs Predicted Values For Kernel Discoloration



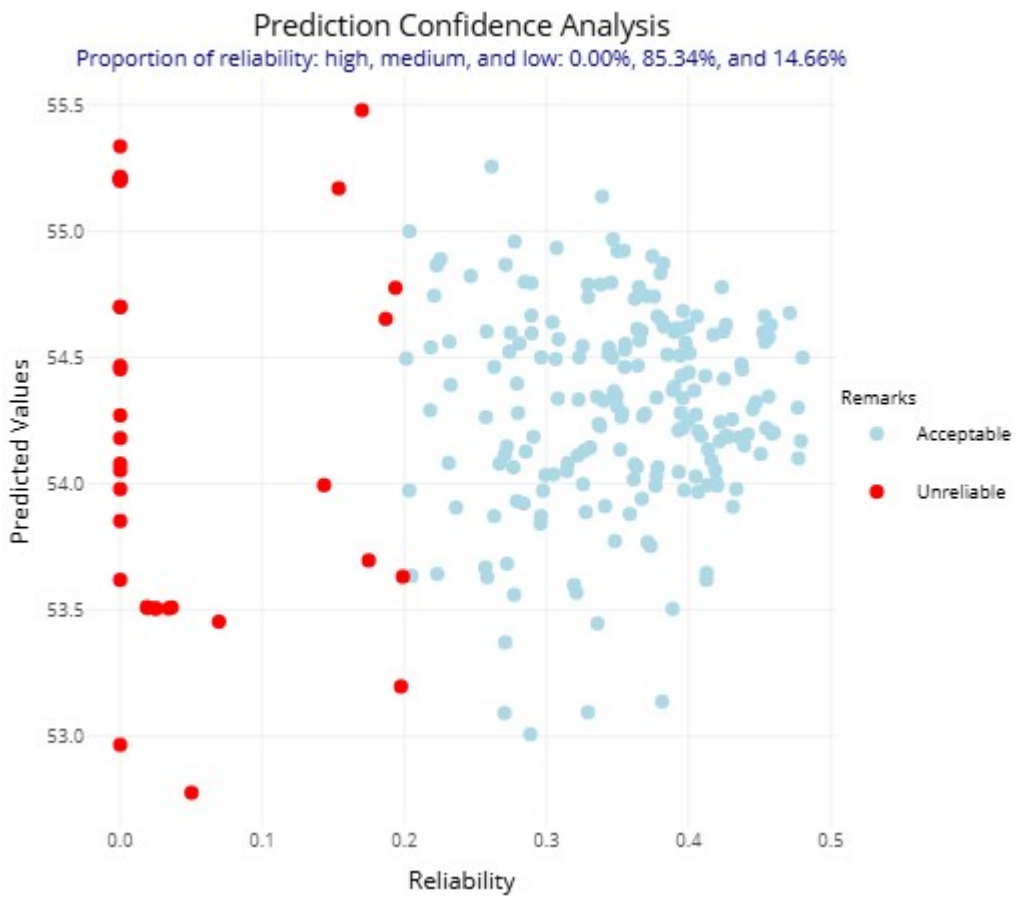


Distribution of Observed vs Predicted Values For Grain Protein Content

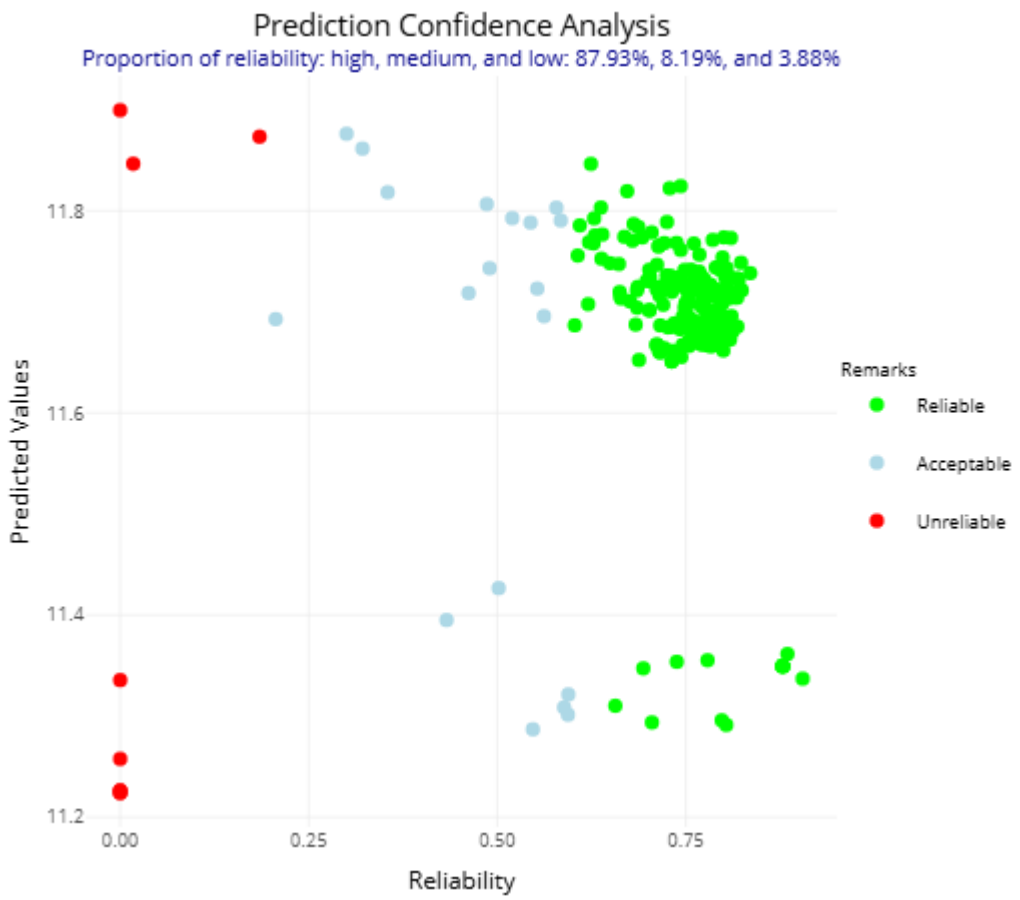


Reliability of Predictions for Kernel Discoloration Resistance and Grain Protein Content

Color:



Protein:



Impacts of Findings

- We showed no correlation between maturity or protein content with kernel discoloration
- There appears to be QTLs controlling grain protein content that do not appear to collocate with the QTLs responsible for kernel discoloration resistance
- Several QTLs were identified across all chromosomes for kernel discoloration resistance

Future Aims of Project

- Field trials from 2025 will be characterized for kernel color and protein
- Develop and validate a genomic selection model for kernel discoloration resistance using more environments

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Cameron.Matthews@ndsu.edu